Cognitive Computing

Hypothesis Generation, Evaluation and Testing

- Question/Problem → Solution/Answer
- Research Problem → Hypotheses → Testable
- Research Problem → Concern or issue that needs to be studied

Difference Between "Solution to a Question" and "Hypothesis to a Research Problem"

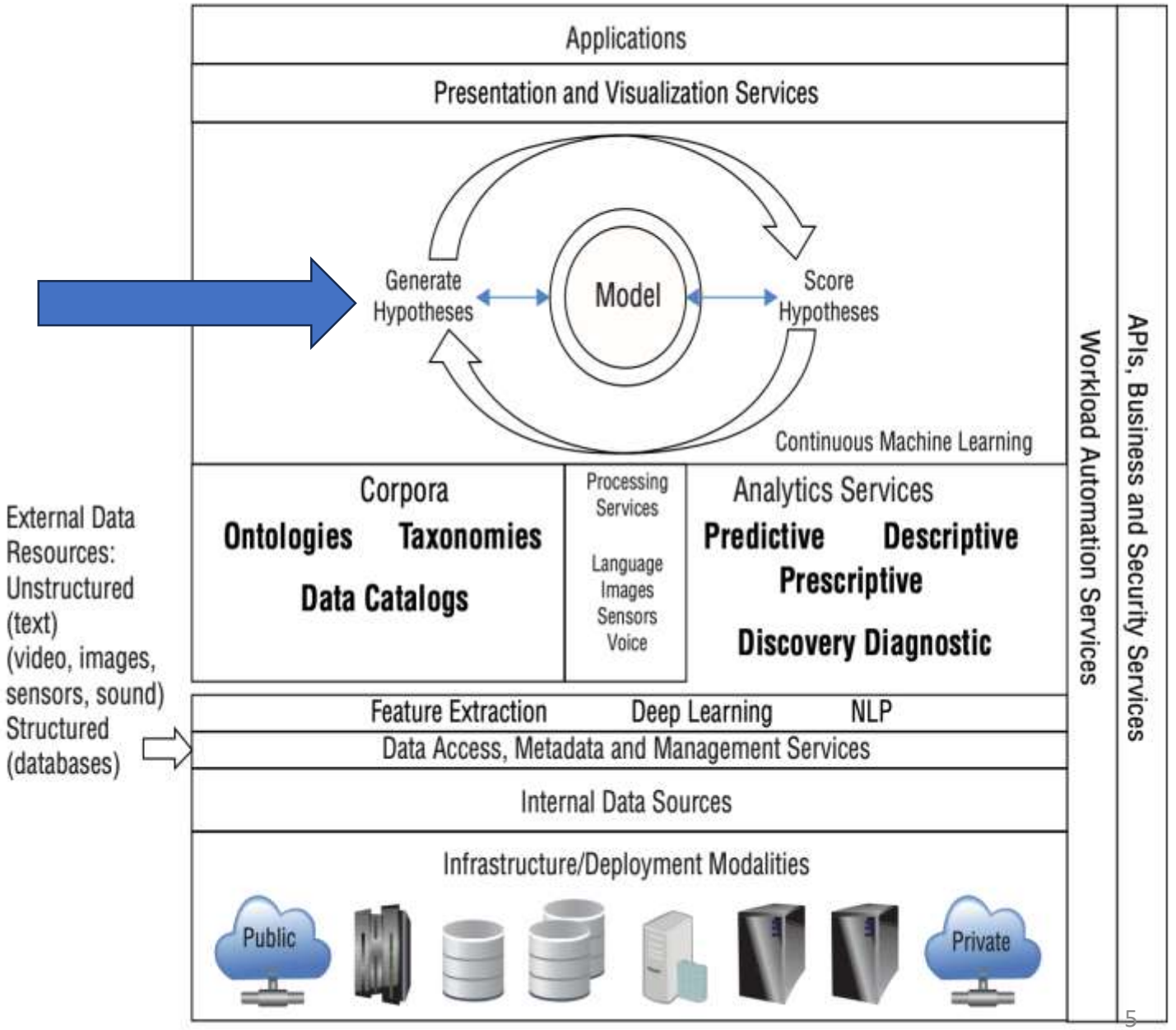
- **Example Question:**
  "What is the best way to reduce blood pressure?"
- **Possible Solution:**
- "Regular exercise, a low-sodium diet, and stress management are effective ways to lower blood pressure."
- **Key Features of a Solution**:
  - **Provides an answer** directly based on existing knowledge or analysis.
  - **Does not require formal testing** but may be backed by existing evidence.
  - **Definitive** rather than probabilistic.
  - Example: "2 + 2 = 4" is a definitive solution, whereas "There is an 80% chance of rain tomorrow" is a probabilistic statement.

- **Example Research Problem:**
  "Does daily exercise reduce blood pressure?"
- **Hypothesis (Testable Statement):**
  - **Null Hypothesis (H_0)**: Daily exercise **does not** reduce blood pressure.
  - **Alternative Hypothesis (H_1)**: Daily exercise **reduces** blood pressure.
- **Key Features of a Hypothesis**:
  - It **proposes** an expected relationship (cause-effect or correlation).
  - It must be **tested using data** (e.g., statistical analysis).
  - The outcome **is not predetermined**; it is either supported or rejected based on evidence.

"Hypothesis to a Research Problem"

- **Key Features of a Hypothesis**:
  - It **proposes** an expected relationship (cause-effect or correlation).
    - Example: Causal hypothesis: "Increased screen time causes reduced sleep quality in teenagers.
    - "Correlational hypothesis: "There is a positive correlation between exercise and mental well-being."
  - It must be **tested using data** (e.g., statistical analysis).
    - Example: Testing whether a new drug reduces blood pressure by comparing results from a control and experimental group.
  - The outcome **is not predetermined**; it is either supported or rejected based on evidence.
    - Example: A study might hypothesize that "A high-protein diet leads to faster muscle growth," but if data shows no significant difference, the hypothesis is rejected.

Hypothesis Generation and Testing (i.e., Scoring)



1. Hypothesis Generation

- Definition- "**A hypothesis may be simply defined as a guess. A scientific hypothesis is an intelligent guess**" - Isaac Asimov
- A *hypothesis* is **a testable assertion based on evidence** that explains some observed phenomenon.
- In a cognitive computing system, you look for evidence to support or refute hypotheses.
- You need to acquire data from various sources, create models, and then test how well the models work.

1. Hypothesis Generation

- If the hypothesis is not supported by the data, the user has several options.
- For example, the user may refine the data by adding to the corpus, or change the hypothesis.
- To evaluate the hypothesis requires a collaborative process of constituents that use the cognitive system.
- Just as with the creation of the hypothesis, the evaluation of results refines those results and trains again.

2. Performing Hypothesis Generation

- The hypothesis generation should be made before looking at the dataset or collection of the data.
- You will notice that if you have done your hypothesis generation adequately, you would have included all the variables present in the dataset in your hypothesis generation.
- You might also have included variables that are not present in the dataset.

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

Let us now look at the “NEW YORK CITY TAXI TRIP DURATION PREDICTION” problem statement and **generate a few hypotheses** that would affect our taxi trip duration to understand hypothesis generation.

Problem-

To predict the duration of a trip so that the company can assign the cabs that are free for the next trip. This will help in reducing the wait time for customers and will also help in earning customer trust.

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

- **Trip Distance-:**

  Hypothesis: Longer trips will generally take more time to complete.

  Rationale: Distance is a direct factor in travel time, assuming no major disruptions like traffic or roadblocks.

- **Passenger Count-:**

  Hypothesis: More passengers may increase trip duration.

  Rationale: More passengers could mean multiple drop-offs, extra stops, or longer boarding times.

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

- **Time of Day**

  Hypothesis: Trips during rush hours will have longer durations.

  Rationale: Morning (7–10 AM) and evening (4–7 PM) traffic congestion slows travel speed.

- **Day of the Week**

  Hypothesis: Weekday and weekend trip durations will differ.

  Rationale: Weekday trips are affected by work commutes, while weekends may see different travel patterns.

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

- **Weather Conditions**

  Hypothesis: Bad weather (rain, snow, fog) will increase trip time.

  Rationale: Poor visibility, slippery roads, and cautious driving reduce travel speed.

- **Pickup and Drop-off Locations**

  Hypothesis: Trips starting/ending in high-traffic areas (e.g., airports, downtown) take longer.

  Rationale: Heavy congestion, passenger queues, and frequent stops can delay travel.

- **Driver Experience**

  Hypothesis: More experienced drivers’ complete trips faster.

  Rationale: Skilled drivers know better routes, shortcuts, and optimal traffic patterns

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

1. After **writing down the hypothesis and looking at the dataset** you will notice that you would have ***covered the writing of hypothesis on most of the features*** present in the data set.
   - Dataset is available - https://www.kaggle.com/c/nyc-taxi-trip-duration/data

Data fields

- **id** - a unique identifier for each trip
- **vendor_id** - a code indicating the provider associated with the trip record
- **pickup_datetime** - date and time when the meter was engaged
- **dropoff_datetime** - date and time when the meter was disengaged
- **passenger_count** - the number of passengers in the vehicle (driver entered value)
- **pickup_longitude** - the longitude where the meter was engaged
- **pickup_latitude** - the latitude where the meter was engaged
- **dropoff_longitude** - the longitude where the meter was disengaged
- **dropoff_latitude** - the latitude where the meter was disengaged
- **store_and_fwd_flag** - This flag indicates whether the trip record was held in vehicle memory before sending to the vendor because the vehicle did not have a connection to the server - Y=store and forward; N=not a store and forward trip
- **trip_duration** - duration of the trip in seconds

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

- **Key Insights:**
- **Most Hypotheses Align with Dataset Features**
  - Many of the hypotheses we generated (e.g., trip distance, passenger count, pickup/drop-off locations, time of day) align well with features present in the dataset.
  - This means we can test our hypotheses using the available data.

Data fields

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Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

2. There could also be a possibility that you might have to work with fewer features and the features on which you have generated hypotheses are not currently being captured/stored by the business and are not available.

- **Key Insights:**
- **Missing or Unavailable Features**
    1. Some features that could significantly impact trip duration (e.g., driver experience, real-time traffic conditions, road closures, weather data) may not be available in the dataset.
    2. If such features are not being captured/stored by the business, we cannot directly test our hypotheses related to them.

Case Study: Hypothesis Generation on “New York City Taxi Trip Duration Prediction”

3. Always go ahead and capture data from external sources if you think that the data is relevant for your prediction but not available in the dataset. Ex: Getting weather information.

Key Insights:

- While the **NYC Taxi Trip Duration dataset** contains many relevant features, some critical factors affecting trip time (e.g., weather, traffic conditions) might be missing. To enhance the model’s accuracy, we should **incorporate external data sources** if they are relevant.
- **Example: Adding Weather Data**
  - Rain, snow, and fog can **increase trip duration** due to slower driving speeds and reduced visibility.
  - Since the dataset does not include weather conditions, we can fetch **historical weather data** for NYC using external APIs.
- **Example: Adding Traffic Data**
  - Real-time or historical congestion levels can be obtained from sources like Google Traffic API or NYC Open Data.

Hypothesis Testing

The system doesn't just propose these candidates and stop — it has to check each one against real data and decide whether the evidence actually supports it, which is precisely what a Z-test or t-test does: given a hypothesis, does the data support rejecting the null, or not?

Hypothesis Testing

- Hypothesis testing is an act in statistics whereby an analyst tests an assumption regarding a population parameter.
- The methodology employed by the analyst depends on the nature of the data used and the reason for the analysis.
- Hypothesis testing is used to assess the plausibility of a hypothesis by using sample data.
  - Such data may come from a larger population, or from a data generating process.
- The word population will be used for both of these cases in the following descriptions.

Population



Which is this data we need for hypothesis testing?

- **Same historical data**, but hypothesis testing and ML training use it for two different jobs at two different points in the pipeline.

Hypothesis testing

- Asks a narrow, confirmatory question about ONE relationship at a time: *"Does rush hour significantly affect trip duration, yes or no, at what confidence level?"* — it's a validation step.



ML training

- Doesn't ask yes/no questions — it fits a function across MANY features simultaneously to directly produce predictions: *"Given distance, time of day, weather, and passenger count all at once, output a predicted duration."*

Hypothesis testing decides WHAT should go into the model.
ML training decides HOW MUCH each of those things matters, all at once.
Same data, two different questions, asked in sequence."

Why a cognitive system needs BOTH, not just one:

- Generation without testing = guessing. A system that proposes "rush hour increases trip duration" and just assumes it's true without checking real data isn't reasoning, it's speculating — this is the difference between a cognitive system and a system that just pattern-matches confidently.
- Testing without generation = nothing to test. Hypothesis testing needs a hypothesis to test in the first place; a system with no generation mechanism has no candidate explanations to evaluate, so it can't even start.

Hypothesis Testing - Example for Cognitive Computing System

- Suppose a healthcare AI system claims that its machine learning model can predict patient readmission rates with 90% accuracy.
- To verify this claim, we conduct hypothesis testing by analyzing past hospital data and evaluating the model’s predictions against actual patient readmission outcomes.
- This helps us determine whether the AI system's claimed accuracy is statistically valid or if there is a significant difference between the expected and observed results.

Hypothesis Testing-Example

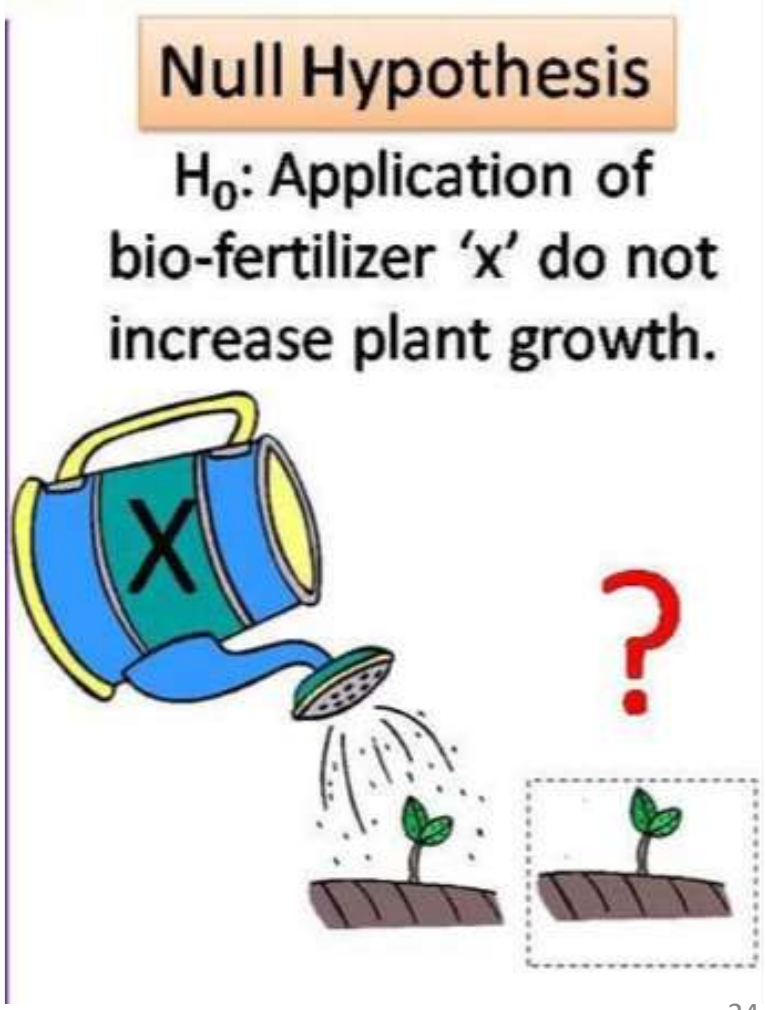
- **An example of hypothesis testing is testing whether the mean weight of apples in a particular fruit garden is 100 grams.**

Types of Hypothesis

- In hypothesis testing, we set up two competing hypotheses:

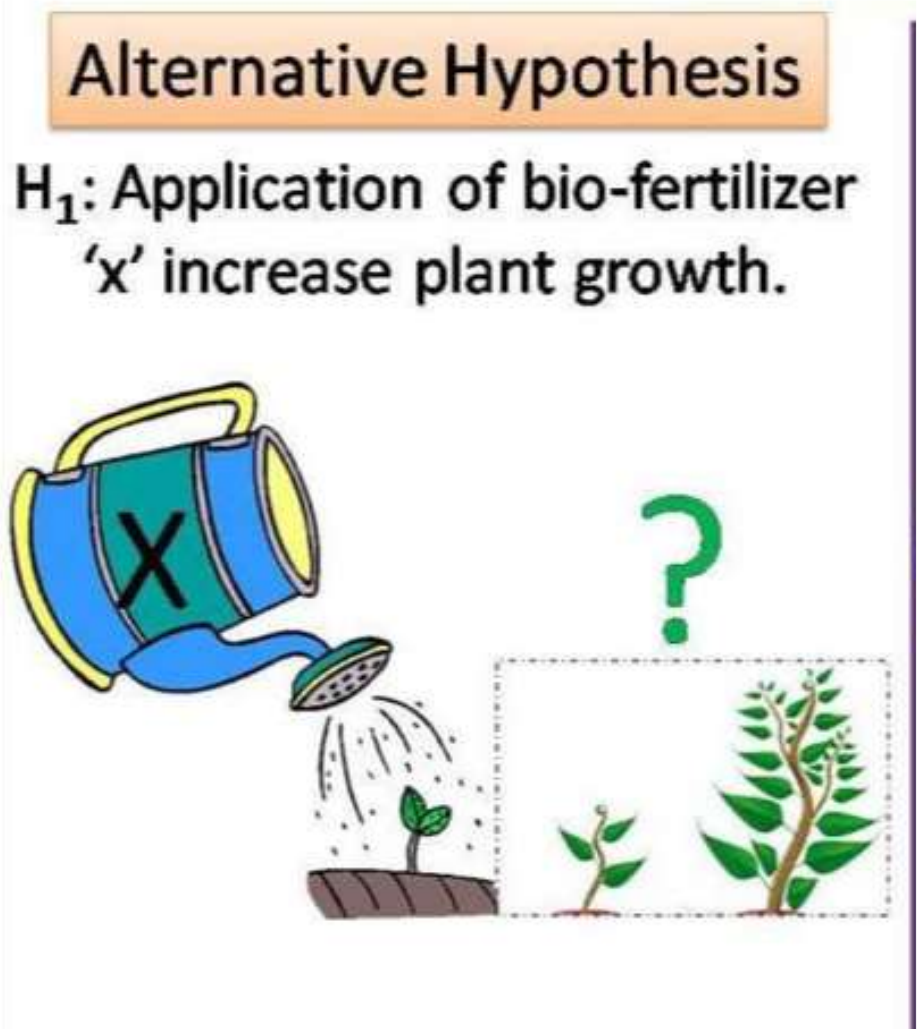
- **Null Hypothesis (H_0)**:
  - Represents the **default assumption**.
  - Assumes that there is **no significant effect or difference**.
  - It is what we **test against** to see if there is enough evidence to reject it.

H_0



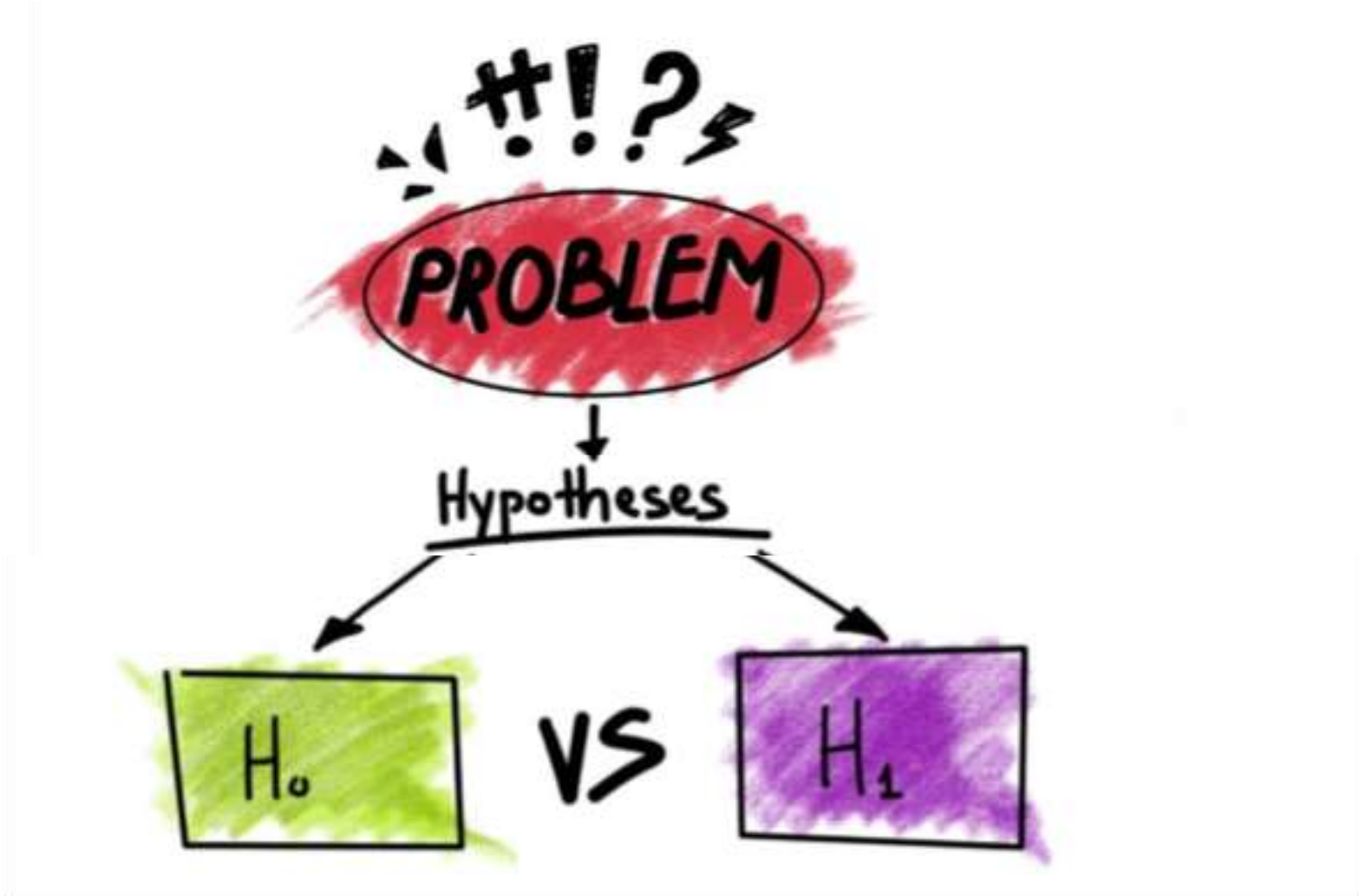
Types of Hypothesis

- In hypothesis testing, we set up two competing hypotheses:



- **Alternative Hypothesis (H_1 or H_a)**:
  - Represents the **research claim** or the **opposite of H_0**.
  - Suggests that there **is a significant effect or difference**.
  - If there is **strong evidence** against H_0, we accept H_1 or H_a.

H_1 or H_a


#!?
PROBLEM
Hypotheses
H₀
VS
H₁

Null Hypothesis

- It is a statement that states there is no significant difference or relationship between two populations or variables.
- The null hypothesis, denoted by $\boldsymbol{H_0}$.
- Example-
  - The null hypothesis is that the mean weight is 100 grams.
    - This means we assume, based on initial expectations, that the true average weight of the population or sample is **100 grams**. In hypothesis testing, we test this assumption using sample data to determine if there is enough evidence to reject it.
  - There is no significant difference between the expected and observed results of AI's system.

Alternative Hypothesis

- It is a statement that there is a significant difference or relationship between the two populations or variables.
- **The alternative hypothesis, denoted by H_1 or H_a.**
- **There are three types of alternative hypotheses. These are:**

  (i) H_1: weight≠100 ,

  (ii) H_1:weight>100 and

  (iii) H_1:weight<100
- We have to consider one of these three hypotheses in a problem.

Alternative Hypothesis

- Example-The alternative hypothesis is that the mean weight is not equal to 100 grams.
- There is a significant difference between the expected and observed results of AI's system.

Decision Errors in Hypothesis Testing

- The two hypotheses are complementary to each other.
- We accept (or reject) the null hypothesis; this is equivalent to rejecting (or accepting) the alternative hypothesis.
- In this process, we make two types of decision errors, namely:
  - Type I error and
  - Type II error

Type I and Type II Error (General Terms)

		Reality: True	Reality: False
Measured	True	Correct (True Positive)	False Positive (Type I error)
	False	False Negative (Type II error)	Correct (True Negative)

Decision Errors in Hypothesis Testing

Measured	Reality: Null Hypothesis is true	Reality: Null Hypothesis is False
Accept Null Hypothesis	Correct Outcome (True Positive) 1 - β	False Negative (Type II error) β
Reject Null Hypothesis	False Positive (Type I error) α	Correct Outcome (True Negative) 1 - α

Decision Errors in Hypothesis Testing

- ***Type I Error-***
  - When we reject a null hypothesis when it is true.
  - The probability of committing a Type I error is called the significance level denoted by α.

- ***Type II Error-***
  - When we accept a null hypothesis when it is false.
  - The probability of committing a Type II error is denoted by β.
  - The probability of not committing a Type II error is called the Power of the test (1-β).

Type-1 and Type-2 Error

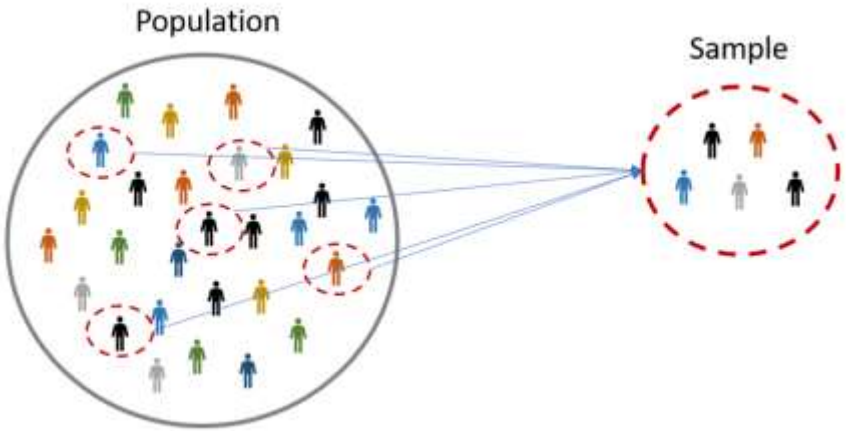
- Example
- Suppose a teacher evaluates the examination paper to decide whether a student passes or fails
  - H_0 -: Student has passed
  - H_1 -: Student has failed
- Type-I error will be the teacher failing the student [rejects H_0 although the student scored the passing marks [H_0 was true]
- Type-II error will be the case where the teacher passes the student [do not reject H_0 although the student did not score the passing marks [H_1 is true]

	Null Hypothesis is true	Null Hypothesis is False
Reject Null Hypothesis	False Positive (Type I error)	Correct Outcome (True Positive)
Accept Null Hypothesis (or fail to Reject)	Correct Outcome (True Negative)	False Negative (Type II error)

Factors Affecting Type I and Type II Error

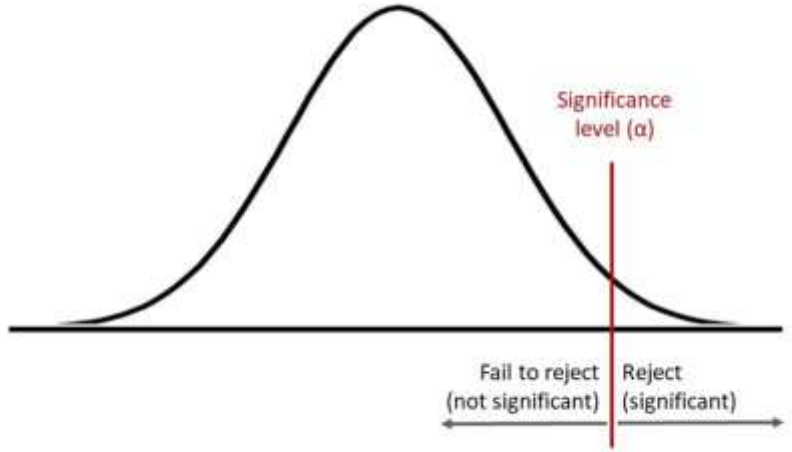
Sample Size

- Larger sample sizes generally reduce the probability of both Type I and Type II errors.
- With larger samples, the estimates tend to be more precise, resulting in more accurate conclusions.



Significance Level (α)

- The significance level (α) in hypothesis testing determines the probability of committing a Type I error.
- Choosing a lower significance level reduces the risk of Type I error but increases the risk of Type II error, and vice versa.
- Most commonly used alpha values are 0.01, 0.05 and 0.1



Level of Significance

- The alpha value is a criterion for determining whether a test statistic is statistically significant
- In a statistical test, Alpha represents an acceptable probability of a Type I error.
- Because alpha is a probability, it can be anywhere between 0 and 1
- In practice, the most commonly used alpha values are 0.01, 0.05 and 0.1 which represent a 1%, 5% and 10% chance of a Type I error, respectively (i.e. rejecting the null hypothesis when it is in fact correct)

Steps in Hypothesis Testing

1. Set up a Hypothesis- State the null and alternative hypotheses.
2. Setup the level of significance- **Significance level is the probability of occurrence of wrong decision. (**If not given then consider 5%.**)**
3. Choose the appropriate test statistic-
    1. Z-Score (sample size is greater than or equal to 30)
    2. t- Test (sample size is less than 30)

} Data that approximately follows a normal distribution

Steps in Hypothesis Testing

4. Computation-

Z-Test-

$$Z = \frac{x - np}{\sqrt{npq}}$$

x-observed success, n-sample size, Z-calculated critical value, p-probability of success, q-probability of failure

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$\bar{x}$ is the sample mean, μ is the population mean, σ is the population standard deviation and n is the sample size.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Steps in Hypothesis Testing

4. Computation-
   - **t-Test-**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

s-standard deviation of the sample, μ - mean of the population, x̄-sample mean, n-sample size

Steps in Hypothesis Testing

5. Making Decision-

$Z_{cal} > Z_{tab}$, hence (H_0) is rejected- For Z-Test

$t_{cal} > t_{tab}$, hence (H_0) is rejected- For t-Test

Hypothesis Testing-Question-1

- A coin was tossed 484 times and Head turned up 265 times. Test the hypothesis that the coin is unbiased. The critical Z-values are given. (Two-tailed test)

p	Significance Level	Confidence Level	Two-*Tailed Test*	One-*Tailed Test*
0.1	10%	90%	1.65	1.28
0.05	5%	95%	1.96	1.64
0.01	1%	99%	2.58	2.33
0.001	0.1 %	99.9%	3.29	3.10

Hypothesis Testing-Solution

- **Step-1 : Set up a Hypothesis- State the null and alternative hypotheses.**
  - H_0 – Coin is unbiased
  - H_a – Coin is biased
- **Step-2: Setup a significance level- If not given then consider 5%.**
- **Step-3: Setting a Test statistic- Z-Score (sample size is greater than 30)**

$$Z = \frac{x - np}{\sqrt{npq}}$$

Hypothesis Testing-Solution

- **Step-4: Calculation**

  n=484, x=265, p=0.5, q=0.5

$$Z = \frac{x - np}{\sqrt{npq}}$$

Put the values:

Z= 265-484*(0.5)/sqrt(484*(0.5)*(0.5)

Z_{cal} =2.09

Z_{tab} = 1.96

Hypothesis Testing- Solution

- **Step-5: Make a decision**
- Check -1: $Z_{cal} > Z_{tab}$, hence (H_0) is rejected.

or

- Check-2: The Z_{cal} (2.09) does not lie in the range of Z_{tab} (i.e. -1.96 to 1.96)

Hence the Null Hypothesis (H_0) is rejected.

Hypothesis Testing- Question-2

- A professor claims that all the students in the first-year class possess above-average IQs. Randomly, a test was conducted on thirty students, resulting in a mean IQ of 117. The population mean was 100, and the standard deviation was 27. (One-tailed Test).

- H_0 – All the students in the first-year class possess above-average IQs
- H_a – All the students in the first-year class do not possess above-average IQs

Setup a significance level- If not given then consider 5%.

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{117 - 100}{\frac{27}{\sqrt{30}}} = \frac{17}{\frac{27}{5.47}} = \frac{92.99}{27} = 3.44$$

Hypothesis Testing-Solution

Solution- one tailed test

$$\mathbf{Z_{Cal}} = 3.44$$

Z_{Tab} =1.64 (Value from the Table given in question 1.)

$Z_{cal} > Z_{tab}$, hence (H_0) is rejected.

Hypothesis Testing- Question-3

The manufacturer of a certain make of Thermometer claims that his thermometers have a mean life of 20 months. A random sample of 7 such thermometers gave the following values. Life of thermometers in months: 19, 21, 25, 16, 17, 14, 21. Can you regard the producer's claim to be valid at 1% level of significance? Given that t_{tab}= 3.707.

19, 21, 25, 16, 17, 14, 21

- Mean of the sample = $\frac{19 + 21 + 25 + 16 + 17 + 14 + 21}{7}$ = 19

- Standard Deviation (s) = $\sqrt{\frac{\sum(x_i - \mu)^2}{N-1}}$

- s = 3.69
- Step-1
  - H_0= Thermometers have a mean life of 20 months.
  - H_a= Thermometers don not have a mean life of 20 months.

Hypothesis Testing-Solution

- μ = 20, n=7, calculate $\bar{x}$ which is the mean of the sample using the data given in the question.
- Then put all the values in the below equation:

$$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{19-20}{\frac{3.69}{\sqrt{7}}} = \frac{-1}{\frac{3.69}{2.64}} = \frac{-2.64}{3.69} = -0.71$$

- t_{cal} = -0.7 and t_{tab}= 3.707
- H_0 is accepted ($t_{cal} < t_{tab}$)

Justify cognitive systems are probabilistic in terms of Hypothesis generation and testing

- Testing a hypothesis never produces certainty — a Z-test tells you whether to reject H_0 at a given confidence level (95%, 99%), not "the absolute truth."
- That's the same probabilistic character as IBM Watson's "flu: 15%" — the system is stating how *confident* it is that a hypothesis holds, not asserting it as fact.

A system randomly guesses that 'driver's favorite color' affects taxi trip duration, collects data, and finds no statistically significant relationship (fails to reject H_0). Did the system fail?

- **No** — this is the loop working correctly.
- A cognitive system is expected to generate implausible hypotheses sometimes; the *testing* step is what filters them out.
- A system that never generates a hypothesis that turns out false isn't being rigorous, it's being timid — the value of Score Hypotheses is precisely that it can say no.